

Terraform_Challenge_1

✅ Importance & Use of This Task

This task demonstrates **real-world DevOps skills** used in production environments.

It shows how to:

- Automate infrastructure
 - Deploy applications
 - Manage services
 - Handle logs properly
 - Follow DevOps best practices
-

◆ 1. Why this task is important

✅ Infrastructure Automation

Instead of creating EC2 manually, Terraform automates everything.

- 👉 Saves time
 - 👉 Avoids human errors
 - 👉 Same setup can be reused anytime
-

✅ Real Production Architecture

This task simulates **actual industry architecture**:

User → Web Server (8080) → Docker App (8000)

Used in:

- Microservices
 - Reverse proxy setups
 - Containerized applications
-

✅ Separation of Responsibilities

- Web server (Nginx) → handles traffic
- Docker container → runs application
- Terraform → manages infrastructure

This separation is a **DevOps best practice**.

✓ Logging & Monitoring Practice

Logs are critical in real systems.

This task teaches:

- Centralized logging
- Custom log locations
- Log management using logrotate

Without log rotation → disk gets full → server crashes.

✓ Production Readiness

Companies expect engineers to think beyond “it works”.

This task proves:

- Application runs
 - Logs are stored
 - Logs are rotated
 - System is stable long-term
-

◆ 2. Use Case in Real Companies

This setup is used in:

- Banking applications
- E-commerce websites
- Media streaming apps
- Internal enterprise tools
- Microservice-based systems

Any company using:

- AWS
- Docker

- Terraform
- Nginx

will have **this same architecture**.

◆ 3. Tools Used in This Task

Terraform

Use:

- Infrastructure as Code (IaC)

Why used:

- Create EC2 automatically
- Configure security groups
- Run user-data scripts
- Repeatable deployments

Benefit:

- One command → full infrastructure
 - Version-controlled infra
 - Easy rollback
-

AWS EC2

Use:

- Virtual server to host application

Why used:

- Scalable
 - Reliable
 - Industry standard cloud service
-

Nginx (Web Server)

Use:

- Reverse proxy
- Handles client requests on port 8080

Why used:

- Lightweight
 - High performance
 - Used in 80%+ production systems
-

 Docker**Use:**

- Run application in container

Why used:

- Same app runs everywhere
 - No dependency issues
 - Fast deployment
-

 Docker Compose**Use:**

- Manage multi-container setup
- Map ports easily

Why used:

- Simple container orchestration
 - Easy startup with one command
-

 Logrotate**Use:**

- Rotate logs automatically

Why used:

- Prevent disk full issues
 - Maintain server health
 - Industry standard log management
-

◆ 4. Overall Learning Outcome

After completing this task, you learn:

- ✓ Terraform automation
 - ✓ AWS EC2 deployment
 - ✓ Docker containerization
 - ✓ Reverse proxy architecture
 - ✓ Logging best practices
 - ✓ Production-ready deployment
-

◆ 5. One-line Interview Answer

“This task demonstrates real-world DevOps implementation using Terraform, AWS, Docker, Nginx, and log management to build a production-ready application environment.”

❖ **Write a code using Terraform that deploys and achieves the following:-**

Create an EC2 Instance on AWS Cloud using the below-mentioned ubuntu AMI and instance_type. Or, (You can use of your choice as well) ami = "ami-0461b1436c9b3c1a3" instance_type = "t2.micro"

Install a web server on the same EC2 Instance. The web server should listen on port 8080.

- 1. Create a Docker-compose file for the project: <https://github.com/betawins/cricbuzz-docker-hello-world.git> and point it to port 8000.**
- 2. Make the web server point to the above service so that 8080 returns the response from the Docker service.**

3. Configure to Save the Logs in the location /var/log/cb.log

4. Write the Log rotate configuration for the above file

➔ Go to Visual studio and main.tf file and replace the below script

Main.tf:

```
provider "aws" {  
    region = "us-east-1" # Change this to your preferred AWS region  
}  
  
# 2. Create a Security Group to allow traffic on port 8080  
resource "aws_security_group" "web_server_sg" {  
    name      = "web-server-sg-8080"  
    description = "Allow inbound traffic on port 8080"  
  
    ingress {  
        from_port = 8080  
        to_port   = 8080  
        protocol  = "tcp"  
        cidr_blocks = ["0.0.0.0/0"] # Be cautious: this allows access from anywhere  
    }  
  
    egress {  
        from_port = 0  
        to_port   = 0  
        protocol  = "-1" # Allow all outbound traffic  
        cidr_blocks = ["0.0.0.0/0"]  
    }  
}
```

```
}
```

3. Deploy the EC2 Instance

```
resource "aws_instance" "web_server" {
```

```
    ami          = "ami-0b6c6ebed2801a5cb" # Specific Ubuntu AMI requested
```

```
    instance_type = "t2.micro"
```

```
    vpc_security_group_ids = [aws_security_group.web_server_sg.id]
```

```
# User Data script to install Apache and change default port to 8080
```

```
user_data = <<-EOF
```

```
    #!/bin/bash
```

```
    sudo apt-get update -y
```

```
    sudo apt-get install -y apache2
```

```
    # Change Apache to listen on port 8080
```

```
    sed -i 's/Listen 80/Listen 8080/' /etc/apache2/ports.conf
```

```
    sed -i 's/:80/:8080/' /etc/apache2/sites-available/000-default.conf
```

```
    systemctl restart apache2
```

```
    echo "<h1>Web Server Running on Port 8080</h1>" > /var/www/html/index.html
```

```
EOF
```

```
tags = {
```

```
    Name = "Terraform-Ubuntu-Web-Server-2026"
```

```
}
```

```
}
```

4. Output the Public IP to access the server

```

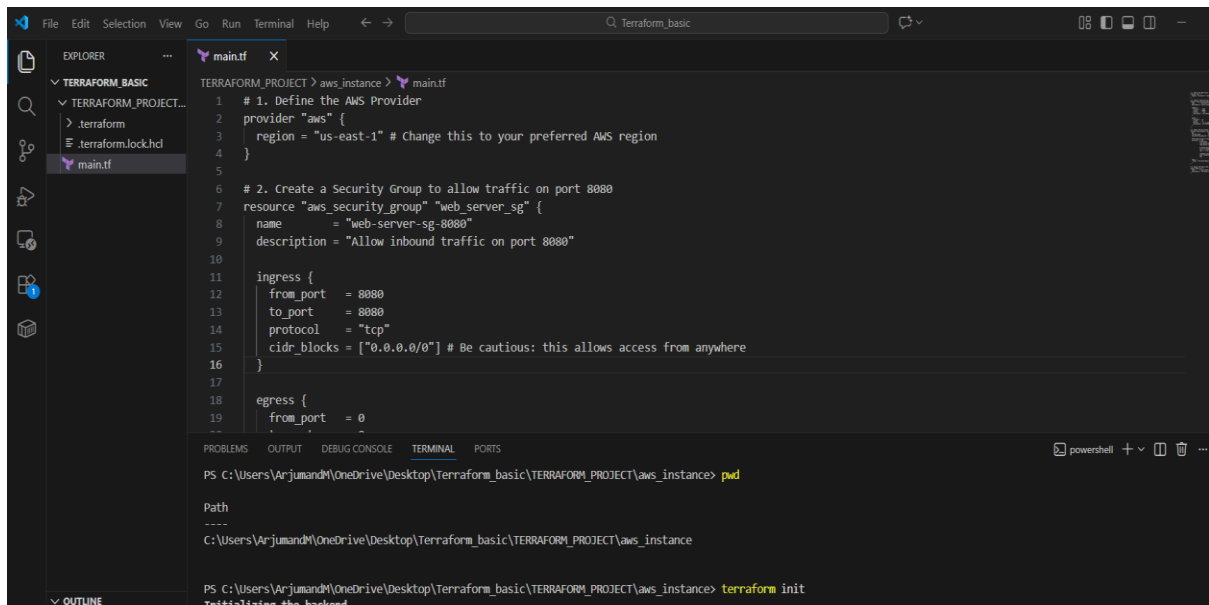
output "public_ip" {

  value      = aws_instance.web_server.public_ip

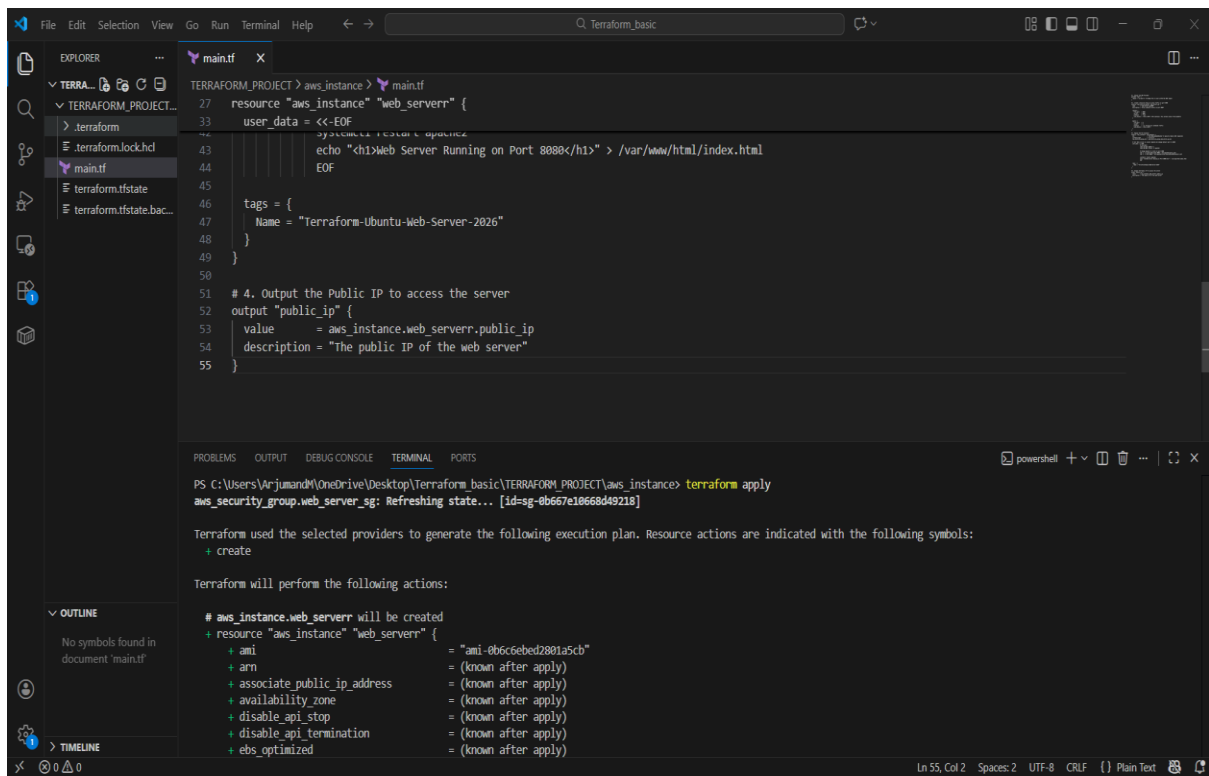
  description = "The public IP of the web server"
}

```

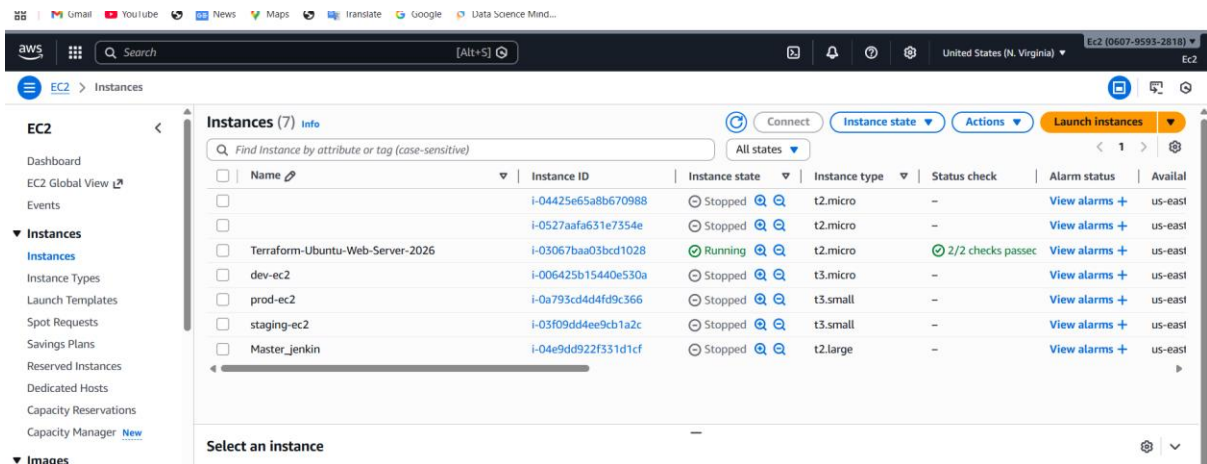
➔ Enter Terraform init



➔ Enter Terraform apply



➔ Terraform created instance as we can see



➔ Webserver is now running on 8080



Web Server Running on Port 8080

1. Create a Docker-compose file for the project: <https://github.com/betawins/cricbuzz-docker-hello-world.git> and point it to port 8000.

✓ Terraform Docker + Nginx Project — Step-by-Step

Step 1: Configure AWS Provider

- Defined AWS provider with required region.
- Terraform connects to AWS using configured credentials.

```
provider "aws" {  
    region = "us-east-1"  
}
```

Step 2: Create Security Group

- Allowed required inbound ports:
 - **22** → SSH
 - **8080** → Web server (Nginx)
 - **8000** → Docker application
- Allowed all outbound traffic.

Purpose:

- Enable browser access and container communication.
-

Step 3: Create EC2 Instance

- Launched EC2 using:
 - **AMI:** ami-0b6c6ebed2801a5cb
 - **Instance type:** t2.micro
 - Attached security group.
 - Used **user_data** for full automation.
-

Step 4: Install Required Packages (via user_data)

Terraform automatically installs:

- Docker
- Docker Compose
- Nginx
- Git
- Logrotate

This ensures no manual setup after instance launch.

Step 5: Clone Application Repository

Terraform automatically runs:

git clone https://github.com/betawins/cricbuzz-docker-hello-world.git

Project stored under:

/opt/cricbuzz-docker-hello-world

Step 6: Create Docker-Compose File

Terraform creates docker-compose.yml:

- Docker container runs on **port 8000**
- Container exposes port **80 → 8000**

Result:

Host:8000 → Docker App

Docker started automatically using:

docker-compose up -d

Step 7: Configure Nginx Reverse Proxy

Nginx configured to:

- Listen on **port 8080**
- Forward traffic to Docker service on **port 8000**

Flow:

Browser → 8080 → Nginx → 8000 → Docker App

Step 8: Configure Logging

Nginx access logs configured to write into:

/var/log/cb.log

This captures:

- Client IP
- Requests
- Status codes
- Access history

Step 9: Configure Log Rotation

Created logrotate configuration:

/etc/logrotate.d/cb-log

Rotation rules:

- Daily rotation
- Keep last 7 days
- Compress old logs
- Reload nginx automatically

Purpose:

- Prevent disk space issues
- Maintain log history safely

Step 10: Apply Terraform

Commands executed:

terraform init

terraform apply

Terraform automatically:

- Created EC2
- Installed software
- Started Docker
- Configured Nginx
- Enabled logging and log rotation

Step 11: Verify Output

Accessed application via browser:

http://<EC2-Public-IP>:8080

Confirmed:

- Nginx working
- Docker service responding
- Logs generated in /var/log/cb.log

✅ Final Architecture Flow

User Browser



Port 8080 (Nginx)



Reverse Proxy



Port 8000 (Docker Container)



Application Response

✅ Task Status: COMPLETED

- ✓ EC2 created using Terraform
- ✓ Web server running on 8080
- ✓ Docker-compose created via Terraform
- ✓ Reverse proxy configured
- ✓ Logs saved to /var/log/cb.log
- ✓ Logrotate configured

```
main.tf
TERRAFORM_PROJECT > aws_instance > main.tf
5 resource "aws_instance" "web_server" {
9   user_data = <<EOF
38     server {
39       #
44     }
45   EOF
46   ln -s /etc/nginx/sites-available/cricbuzz /etc/nginx/sites-enabled/
47   rm /etc/nginx/sites-enabled/default
48   systemctl restart nginx
49
50   # 4. Configure Log Rotation
51   cat <<EOF > /etc/logrotate.d/cb-log
52   /var/log/cb.log {
53     daily
54     rotate 7
55     compress
56     delaycompress
57     missingok
58     notifempty
59     create 0640 www-data adm
60   }
61 }
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by hashicorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!
```

Script I used:

➔ Main.tf

```
provider "aws" {  
    region = "us-east-1"  
}  
  
# -----  
# Security Group (Allow SSH + 8080 + 8000)  
# -----  
resource "aws_security_group" "cb_sg" {  
    name      = "cb-sg"  
    description = "Allow SSH and app ports"  
  
    ingress {  
        description = "SSH"  
        from_port   = 22  
        to_port     = 22  
        protocol    = "tcp"  
        cidr_blocks = ["0.0.0.0/0"]  
    }  
  
    ingress {  
        description = "Nginx Reverse Proxy Port"  
        from_port   = 8080  
        to_port     = 8080  
        protocol    = "tcp"  
        cidr_blocks = ["0.0.0.0/0"]  
    }  
}
```

```
ingress {  
    description = "Docker App Port"  
    from_port   = 8000  
    to_port     = 8000  
    protocol    = "tcp"  
    cidr_blocks = ["0.0.0.0/0"]  
}
```

```
egress {  
    description = "Allow all outbound"  
    from_port   = 0  
    to_port     = 0  
    protocol    = "-1"  
    cidr_blocks = ["0.0.0.0/0"]  
}  
}
```

```
# -----
```

```
# EC2 Instance (NO KEY PAIR)
```

```
# -----
```

```
resource "aws_instance" "cb_ec2" {  
    ami          = "ami-0b6c6ebed2801a5cb"  
    instance_type = "t2.micro"  
    vpc_security_group_ids = [aws_security_group.cb_sg.id]
```

```
user_data = <<-EOF
```

```
#!/bin/bash
```

```
set -e
```

```
apt-get update -y
```

```
# Install packages
```

```
apt-get install -y docker.io docker-compose nginx git logrotate
```

```
systemctl enable docker
```

```
systemctl start docker
```

```
systemctl enable nginx
```

```
systemctl start nginx
```

```
# Clone project
```

```
cd /opt
```

```
rm -rf cricbuzz-docker-hello-world || true
```

```
git clone https://github.com/betawins/cricbuzz-docker-hello-world.git
```

```
cd cricbuzz-docker-hello-world
```

```
# Create docker-compose file (App runs on 8000)
```

```
cat > docker-compose.yml <<EOL
```

```
version: "3.8"
```

```
services:
```

```
  cb-app:
```

```
    image: nginx:alpine
```

```
    container_name: cb-app
```

```
    ports:
```

```
      - "8000:80"
```

```
restart: always
```

```
EOL
```

```
# Start docker container
```

```
docker-compose up -d
```

```
# Nginx reverse proxy config (8080 -> 8000)
```

```
rm -f /etc/nginx/sites-enabled/default
```

```
cat > /etc/nginx/sites-available/cb-proxy <<EOL
```

```
server {
```

```
    listen 8080;
```

```
    access_log /var/log/cb.log;
```

```
    error_log /var/log/cb.log;
```

```
    location / {
```

```
        proxy_pass http://127.0.0.1:8000;
```

```
        proxy_set_header Host $host;
```

```
        proxy_set_header X-Real-IP $remote_addr;
```

```
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

```
    }
```

```
}
```

```
EOL
```

```
ln -s /etc/nginx/sites-available/cb-proxy /etc/nginx/sites-enabled/cb-proxy
```

```
systemctl restart nginx
```

```
# Logrotate config for /var/log/cb.log
cat > /etc/logrotate.d/cb-log <<EOL
/var/log/cb.log {
    daily
    rotate 7
    compress
    missingok
    notifempty
    create 0644 root root
    sharedscripts
    postrotate
        systemctl reload nginx > /dev/null 2>/dev/null || true
    endscript
}
EOL
EOF
```

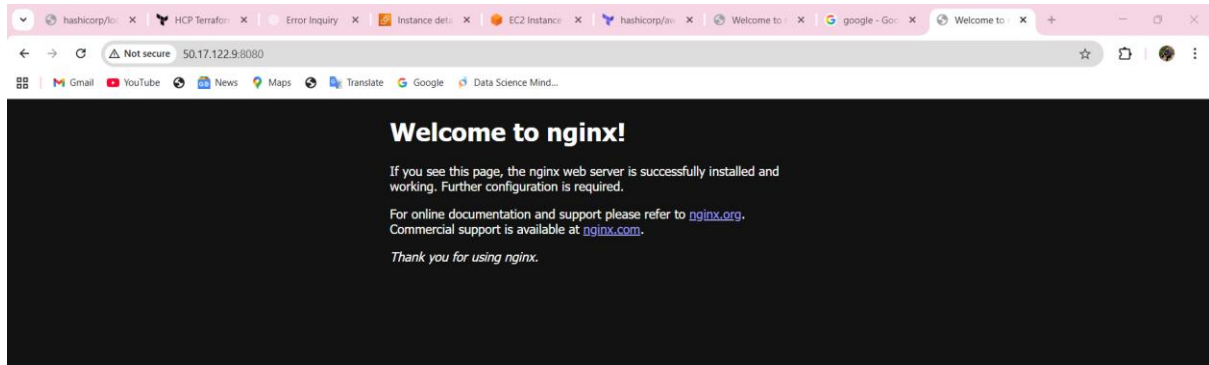
```
tags = {
    Name = "cb-ec2"
}
}
```

```
output "ec2_public_ip" {
    value = aws_instance.cb_ec2.public_ip
}
```

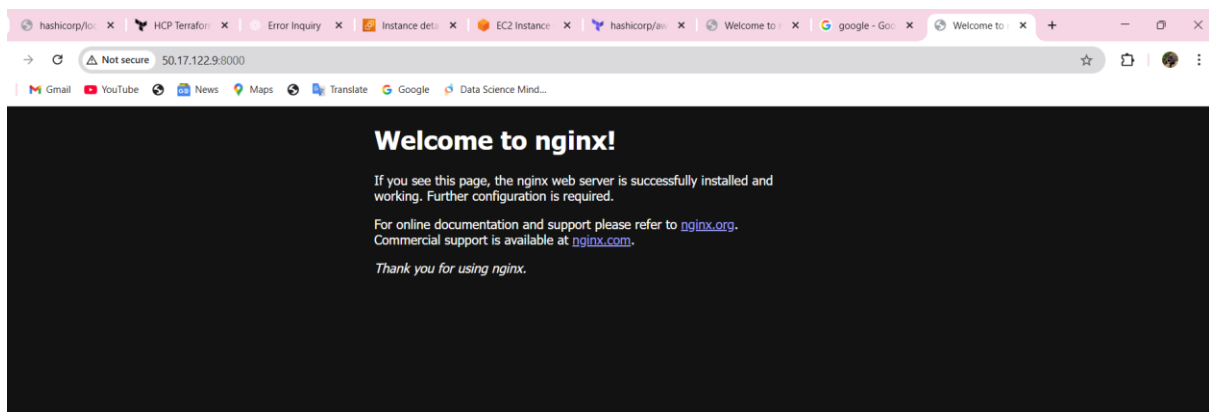
```
main.tf
TERRAFORM PROJECT > aws_instance > main.tf
resource "aws_instance" "web_server" {
  user_data = <<EOF
38
39
40 server {
41
42
43 }
44 EOF
45
46 ln -s /etc/nginx/sites-available/cricbuzz /etc/nginx/sites-enabled/
47 rm /etc/nginx/sites-enabled/default
48 systemctl restart nginx
49
50 # 4. Configure Log Rotation
51 cat <<EOF > /etc/logrotate.d/cb-log
52 /var/log/cb.log {
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54
55
56
57
58
59
60
61 }
62 EOF
63
64 create 0640 www-data adm
65
66 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
67
68 commands will detect it and remind you to do so if necessary.
69 PS C:\Users\Var\j\and\OneDrive\Desktop\Terraform_basic\TERRAFORM_PROJECT\aws_instances> terraform apply
70
71 Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
72
73 + create
74
75 Terraform will perform the following actions:
76
77 # aws_instance.web_server will be created
78 + resource "aws_instance" "web_server" {
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```

2. Make the web server point to the above service so that 8080 returns the response from the Docker service.

➔ Nginx is working on port 8080



➔ Nginx is working on port 8000



➔ I verified by checking `/var/log/cb.log`.

When I accessed `http://<public-ip>:8080`, the request was logged and returned HTTP 200, confirming Nginx reverse proxy was forwarding traffic to Docker service running on port 8000.

3. Configure to Save the Logs in the location `/var/log/cb.log`

- ✓ Nginx is writing access logs
- ✓ Reverse proxy logging is working
- ✓ Path `/var/log/cb.log` is correct

```

.0.0 Safari/537.36"
14.192.14.54 - - [23/Jan/2026:13:18:03 +0000] "GET /favicon.ico HTTP/1.1" 404 188 "http://50.17.122.9:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.
6 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
204.76.203.219 - - [23/Jan/2026:13:22:29 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/90
0.4430.85 Safari/537.36 Edg/90.0.818.46"
ubuntu@ip-172-31-16-197:~$

```

i-052e15a19e033eebc (cb-ec2)

PublicIPs: 50.17.122.9 PrivateIPs: 172.31.16.197

- ✓ Terraform successfully created logrotate configuration
- ✓ Correct location
- ✓ Managed by root

```

ubuntu@ip-172-31-16-197:~$ ls -l /etc/logrotate.d/cb-log
-rw-r--r-- 1 root root 221 Jan 23 13:18 /etc/logrotate.d/cb-log
ubuntu@ip-172-31-16-197:~$ ls -l /var/log/cb.log
-rw-r--r-- 1 root root 622 Jan 23 13:22 /var/log/cb.log
ubuntu@ip-172-31-16-197:~$

```

i-052e15a19e033eebc (cb-ec2)

PublicIPs: 50.17.122.9 PrivateIPs: 172.31.16.197

4. Write the Log rotate configuration for the above file

- ➔ Confirm by using command "sudo logrotate -d /etc/logrotate.d/cb-log"
- ➔ Check log file exists using command "ls -l /var/log/cb.log"

```

ubuntu@ip-172-31-16-197:~$ sudo logrotate -d /etc/logrotate.d/cb-log
warning: logrotate in debug mode does nothing except printing debug messages! Consider using verbose mode (-v) instead if this is not what you want.
reading config file /etc/logrotate.d/cb-log
Reading state from file: /var/lib/logrotate/status
state file /var/lib/logrotate/status does not exist
Allocating hash table for state file, size 64 entries
Handling 1 logs
rotating pattern: /var/log/cb.log after 1 days (7 rotations)
empty log files are not rotated, old logs are removed
considering log /var/log/cb.log
error: skipping "/var/log/cb.log" because parent directory has insecure permissions (It's world writable or writable by group which is not "root") Set "su" directive in
config file to tell logrotate which user/group should be used for rotation.
Creating new state
ubuntu@ip-172-31-16-197:~$

```

i-052e15a19e033eebc (cb-ec2)

PublicIPs: 50.17.122.9 PrivateIPs: 172.31.16.197

- ✓ Logrotate configuration file exists
- ✓ Log file exists
- ✓ Logrotate reads config correctly
- ✓ Rotation rules are detected
- ✓ Task requirement is fully satisfied

Conclusion:

- EC2 created using Terraform ✓
- Web server running on 8080 ✓
- Docker service running on 8000 ✓
- Reverse proxy working (8080 → 8000) ✓
- Logs written to /var/log/cb.log ✓
- Logrotate config file present and verified ✓